

Form 5

HKCEE 1989 Mathematics II

Label 1 $3^{n-1} \times 3^{n+1}$

- A. 3^{n^2-1}
- B. 9^{n^2-1}
- C. 3^{2n}
- D. 6^{2n}
- E. 9^{2n}

Label 2 $\frac{27x^3-8}{3x-2} =$

- A. $(3x-2)^2$
- B. $9x^2-4$
- C. $9x^2+4$
- D. $9x^2-6x+4$
- E. $9x^2+6x+4$

Label 3 $\sqrt{\frac{x}{\sqrt{x}}} =$

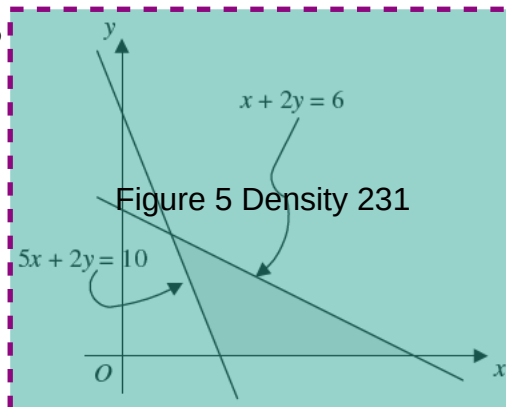
- A. $x^{\frac{3}{4}}$
- B. $x^{\frac{1}{4}}$
- C. $x^{\frac{1}{2}}$
- D. $x^{-\frac{1}{4}}$
- E. $x^{-\frac{3}{4}}$

Label 4 If $f(x) = \frac{x}{1-x}$, then $f(\frac{1}{x}) =$

- A. $\frac{1}{x-1}$
- B. $\frac{1}{1-x}$
- C. $\frac{x}{x-1}$
- D. $\frac{x}{1-x}$

Label 5 $\frac{11-x}{x}$

Label 5



Which of the following systems of inequalities is represented by the shaded region in the figure?

- A. $\begin{cases} x+2y \geq 6 \\ 5x+2y \geq 10 \\ y \geq 0 \end{cases}$
- B. $\begin{cases} x+2y \leq 6 \\ 5x+2y \leq 10 \\ x \geq 0 \end{cases}$
- C. $\begin{cases} x+2y \geq 6 \\ 5x+2y \leq 10 \\ x \geq 0 \end{cases}$
- D. $\begin{cases} x+2y \leq 6 \\ 5x+2y \geq 10 \\ y \geq 0 \end{cases}$
- E. $\begin{cases} x+2y \geq 6 \\ 5x+2y \leq 10 \\ y \geq 0 \end{cases}$

Label 6 Let $f(x) = ax^2 - 5$ and $g(x) = 27x^3 - 18x + 4$. If both expressions leave the same remainder when divided by $3x + 1$, then $a =$

- A. -74
- B. 0
- C. 36
- D. 76
- E. 126

Label 7 If $3x > -2y$ and $y < 0$, then

- A. $\frac{x}{y} > -\frac{3}{2}$
- B. $\frac{x}{y} > \frac{2}{3}$
- C. $\frac{x}{y} < \frac{2}{3}$
- D. $\frac{x}{y} > -\frac{2}{3}$
- E. $\frac{x}{y} < -\frac{2}{3}$

Label 8 Given that r is the only real root of $x^5 + x - 1 = 0$, which of the following ranges contains r ?

- A. $-2 < r < -1$
- B. $-1 < r < 0$
- C. $0 < r < 1$
- D. $1 < r < 2$
- E. $2 < r < 3$

Label 9 If z varies inversely as x and directly as y , then

- A. xyz is a constant
- B. $\frac{xz}{y}$ is a constant
- C. $\frac{yz}{x}$ is a constant
- D. $\frac{xz^2}{y}$ is a constant
- E. $\frac{z^2}{xy}$ is a constant

Label 10 Which of the following is/are true?

- I. If both 2 and 3 are factors of m , then 6 is also a factor of m .
- II. If 15 is a factor of n , then both 3 and 5 are factors of n .
- III. If p is a multiple of both 4 and 6, then p is also a multiple of 24.

- A. I only
- B. II only
- C. I and II only
- D. II and III only
- E. I, II and III

Label 11

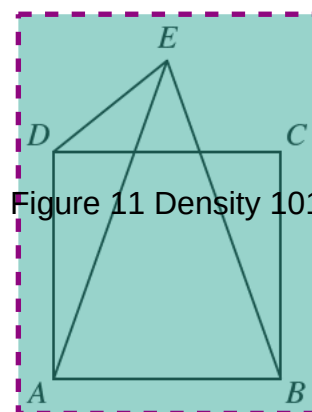


Figure 11 Density 101

In the figure, $ABCD$ is a square and $AE = BE$. $\frac{\text{Area of } AED}{\text{Area of } ABCD} =$

- A. $\frac{1}{2}$
- B. $\frac{3}{8}$
- C. $\frac{1}{3}$
- D. $\frac{1}{4}$
- E. $\frac{1}{8}$

Label 12

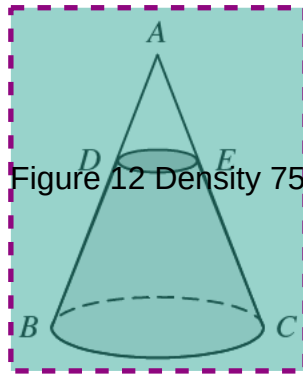


Figure 12 Density 75

A right conical vessel placed on horizontal ground contains some water as shown in the figure. If $AD : DB = 2 : 3$, then $\frac{\text{volume of empty space}}{\text{volume of water}} =$

- A. $\frac{4}{9}$
- B. $\frac{8}{19}$
- C. $\frac{8}{27}$
- D. $\frac{8}{117}$
- E. $\frac{8}{125}$

Label 13 If A is greater than B by 20% and B is smaller than C by 30%, then

- A. A is smaller than C by 16%
- B. A is smaller than C by 6%
- C. A is greater than C by 6%
- D. A is greater than C by 10%
- E. A is greater than C by 16%

Label 14 At the beginning of a year, a man borrows \$1000 from a bank at 5% per annum, compounded yearly. He promises to repay \$300 at the end of each year. How much will he still owe the bank just after the second repayment?

- A. \$402.5
- B. \$450
- C. \$487.5

- D. \$500
- E. \$502.5

Label 15 The least value of $9\cos^2\theta - 6\cos\theta + 1$ is

- A. -4
- B. 0
- C. 1
- D. 4
- E. 16

Label 16 $\frac{1}{\frac{1}{\cos\theta} - 1} - \frac{1}{\frac{1}{\cos\theta} + 1} =$

- A. $\frac{2}{\tan^2\theta}$
- B. $\frac{2}{\tan\theta}$
- C. $2\tan^2\theta$
- D. $\frac{2\cos\theta}{\sin^2\theta}$
- E. $\frac{2\cos^2\theta}{\sin\theta}$

Label 17

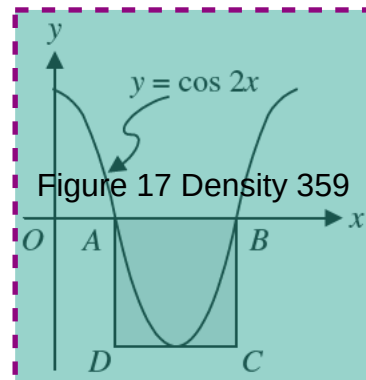


Figure 17 Density 359

The figure shows the graph of $y = \cos 2x$, where $0 \leq x \leq \pi$. The area of the rectangle $ABCD$ is

- A. $\frac{\pi}{2}$
- B. $\frac{\pi}{4}$
- C. π

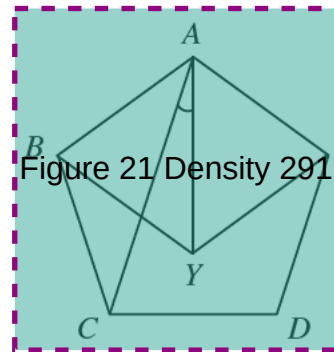
- D. $\frac{3\pi}{2}$
E. 2π

- C. 40
D. 50
E. 80

Label 18 Given that $0^\circ \leq \theta \leq 180^\circ$, how many roots has the equation $(\sin \theta + 1)(\tan \theta + 3) = 0$?

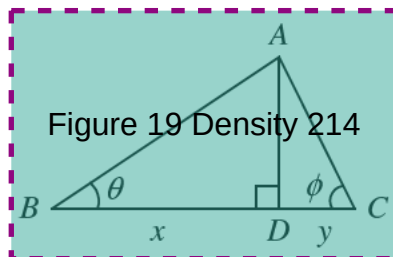
- A. 0
B. 1
C. 2
D. 3
E. 4

Label 21



E

Label 19



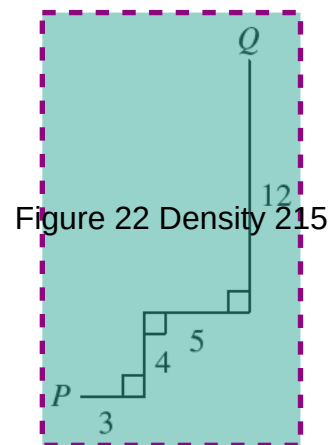
In the figure, $AD \perp BC$. Find $\frac{x}{y}$.

- A. $\frac{\sin \phi}{\sin \theta}$
B. $\frac{\cos \phi}{\cos \theta}$
C. $\frac{\tan \phi}{\tan \theta}$
D. $\frac{\cos \theta}{\cos \phi}$
E. $\frac{\tan \theta}{\tan \phi}$

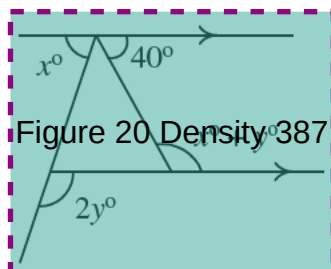
In the figure, $ABCDE$ is a regular pentagon and $ABYE$ is a rhombus. Find $\angle CAE$.

- A. 27°
B. 24°
C. 21°
D. 18°
E. 15°

Label 22



Label 20



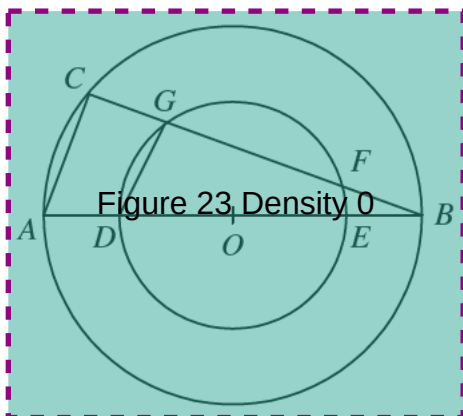
Referring to the figure, find y .

- A. 20
B. 30

Referring to the figure, find the length of the line segment joining P and Q .

- A. 25
B. $10\sqrt{5}$
C. 18
D. $8\sqrt{5}$
E. $\sqrt{194}$

Label 23

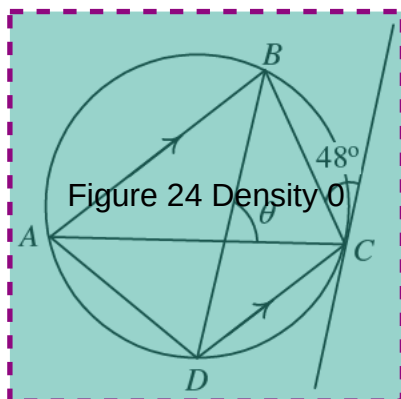


In the figure O is the centre of two Concentric circles. $ADOEB$ and $CGFB$ are straight lines. Which of the following is/are true?

- I. $AC \parallel DG$
- II. $BF = CG$
- III. A, E, F and C are concyclic

- A. I only
- B. II only
- C. I and II only
- D. I and III only
- E. I, II and III

Label 24



In the figure, TC is a tangent to the circle at C and $AB \parallel DC$. If $\angle BCT = 48^\circ$, then $\theta =$

- A. 48°
- B. 72°
- C. 84°
- D. 90°
- E. 96°

Label 25 Referring to the data 1, 1, 1, 1, 1, 2, 2, 2, 3, which of the following is/are true?

- I. median $<$ mean
- II. range $= 3$
- III. mode $= 3$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I, II and III

Label 26 A **BIASED** die is thrown. Suppose the probabilities of getting 1, 2, 3, 4, and 4 are respectively $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$ and $\frac{1}{32}$.

What is the probability of getting 6?

- A. $\frac{1}{64}$
- B. $\frac{1}{36}$
- C. $\frac{1}{32}$
- D. $\frac{1}{12}$
- E. $\frac{1}{6}$

Label 27 A bag contains 4 red, 3 green and 2 white balls. Three men A, B and C each draw one ball in turn from the bag at random without replacement. If A draws first, B second and C third, what is the probability that the balls drawn by B and C are both white?

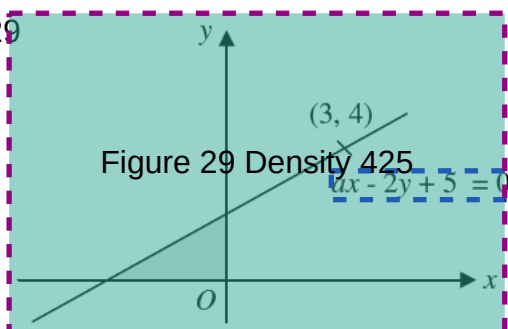
- A. $\frac{1}{36}$
- B. $\frac{1}{28}$
- C. $\frac{4}{81}$
- D. $\frac{25}{72}$

E. $\frac{11}{28}$

Label 28
28. The equation of the straight line perpendicular to $2x + y - 3 = 0$ and passing through $(1, -1)$ is

- A. $x + 2y + 1 = 0$.
B. $x - 2y - 3 = 0$.
C. $-x + 2y - 1 = 0$.
D. $2x + y - 1 = 0$.
E. $2x - y - 3 = 0$.

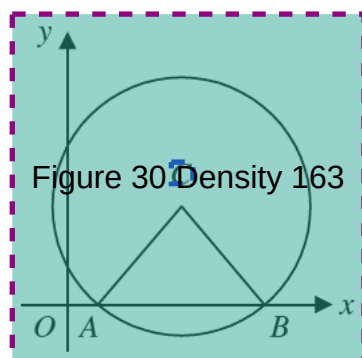
Label 29
29.



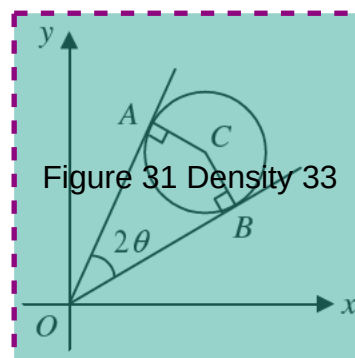
In the figure, the line $ax - 2y + 5 = 0$ passes through the point $(3, 4)$. What is the area of the shaded part?

- A. $\frac{6}{5}$
B. $\frac{25}{4}$
C. $\frac{10}{3}$
D. $\frac{12}{5}$
E. $\frac{25}{2}$

Label 30
30.



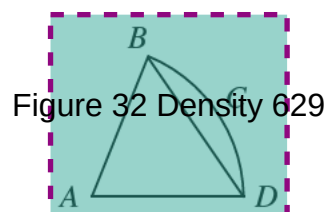
Label 31
31.



In the figure, C is the centre of the circle $x^2 + y^2 - 6x - 8y + 21 = 0$. OA and OB are tangents. If $\angle AOB = 2\theta$, find $\sin \theta$.

- A. $\frac{\sqrt{21}}{5}$
B. $\frac{4}{5}$
C. $\frac{3}{5}$
D. $\frac{2}{\sqrt{21}}$
E. $\frac{2}{5}$

Label 32
32.



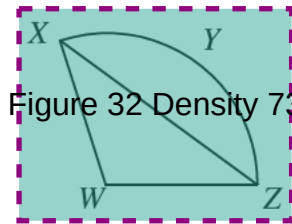


Figure 32 Density 73

In the figure, $ABCD$ and $WXYZ$ are sectors of equal radii. If arc $BCD : \text{arc } XYZ = s : t$, then which of the following is/are true?

- I. $\frac{BD}{XZ} = \frac{s}{t}$
- II. $\frac{\text{area of sector } ABCD}{\text{area of sector } WXYZ} = \frac{s}{t}$
- III. $\frac{\angle BAD}{\angle XWZ} = \frac{s}{t}$

- A. I only
- B. II only
- C. III only
- D. I and III only
- E. II and III only

Label 33

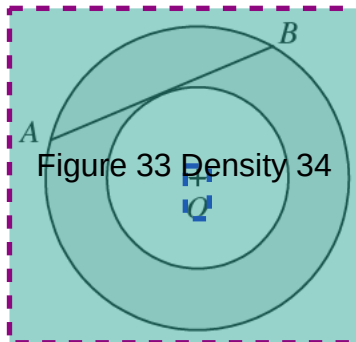


Figure 33 Density 34

In the figure, O is the centre of two concentric circles. AB is tangent to the smaller circle. If $AB = 2$, find the area of the shaded part.

- A. π
- B. $\frac{\pi}{2}$
- C. 2π
- D. 4π
- E. It cannot be found.

Label 34 If 10 arithmetic means are inserted between a and b , then the last one is

- A. $\frac{10a+b}{11}$
- B. $\frac{9a+b}{10}$
- C. $\frac{10(b-a)}{11}$
- D. $\frac{a+9b}{10}$
- E. $\frac{a+10b}{11}$

Label 35 Given that $y \propto \frac{1}{x}$, if x increased by 25%, find the percentage change in y .

- A. Decreased by 20%
- B. Decreased by 25%
- C. Decreased by 80%
- D. Increased by 20%
- E. Increased by 25%

Label 36 The costs of two kinds of coffee A and B are \$12/kg and \$20/kg respectively. In what ratio by weight should A and B be mixed so that the mixture will cost \$15/kg?

- A. 4 : 3
- B. 5 : 2
- C. 5 : 3
- D. 3 : 2
- E. 5 : 4

Label 37

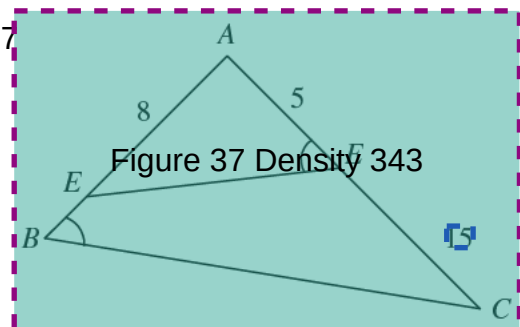


Figure 37 Density 343

In the figure, D and E are points on AB and AC respectively such that $\angle ABC = \angle AED$, $AD = 8$, $AE = 5$ and $EC = 15$. If the area of $\triangle ADE$ is 16, then the area of the quadrilateral $BCED$ is

- A. 200
- B. 100
- C. 96
- D. 84
- E. 40

Label 38

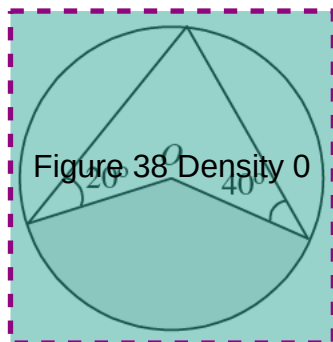


Figure 38 Density 0

In the figure, O is the centre of the circle of radius 6 cm. The area of the shaded part is

- A. $2\pi \text{ cm}^2$
- B. $4\pi \text{ cm}^2$
- C. $6\pi \text{ cm}^2$
- D. $9\pi \text{ cm}^2$
- E. $12\pi \text{ cm}^2$

Label 39 If the sum to infinity of the G.P. $1, -t,$

$t^2, -t^3, \dots$ is $\frac{2}{3}$, find the fourth term.

- A. $\frac{1}{16}$
- B. $\frac{1}{8}$
- C. $\frac{1}{16}$
- D. $\frac{1}{8}$
- E. $\frac{5}{8}$

Label 40 If $\frac{x+3y}{2x+y} = 2$, find $\frac{3x+y}{x+2y}$

- A. 2
- B. 3
- C. 1
- D. 1
- E. 6

Label 41 $\frac{(1-x^2)^n + (1-x)^n}{(1-x)^{2n}}$

- A. $\frac{(1+x)^n + 1}{(1-x)^n}$
- B. $\frac{2-x-x^2}{(1-x)^2}$
- C. $\frac{(1+x)^n + 1}{(1-x)^2}$
- D. $\frac{(1-x)^n + 1}{(1+x)^n}$
- E. $\frac{2-x^n+x^{2n}}{1-x^{2n}}$

Label 42 $\log_4 2\sqrt{2} =$

- A. 3
- B. 3
- C. 1
- D. 2
- E. 2

Label 43 If $x = \sqrt{a+1} - \sqrt{a}$, where $a > 0$, then

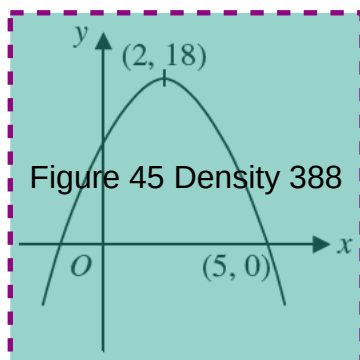
- A. $\frac{1}{x}$
- B. $x + \frac{1}{x}$

- A. 2
 B. $2\sqrt{a}$
 C. $2\sqrt{a+1}$
 D. $2\sqrt{a+1} - \sqrt{a}$
 E. $2(\sqrt{a+1} + \sqrt{a})$

Label 44
 44 If p is a root of $ax^2 + bx + c = 0$, which of the following is a root of $a\left(\frac{x-3}{2}\right)^2 + b\left(\frac{x-3}{2}\right) + c = 0$?

- A. $2p + 3$
 B. $2p - 3$
 C. $3 - 2p$
 D. $\frac{p+3}{2}$
 E. $\frac{p-3}{2}$

Label 45
 45



In figure shows the graph of a quadratic function $y = f(x)$. Given that the graph has vertex $(2, 18)$ and it cuts the x -axis at $(5, 0)$, find the quadratic function.

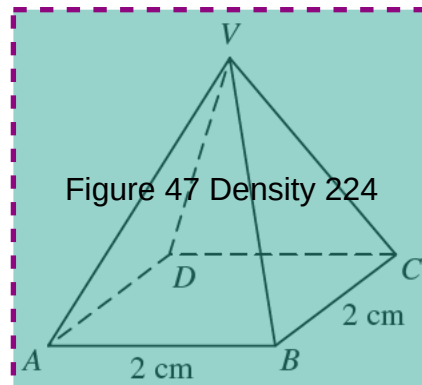
- A. $y = (x - 2)^2 + 18$
 B. $y = -(x - 2)^2 + 18$
 C. $y = (x + 1)(x - 5)$
 D. $y = -2(x + 1)(x - 5)$
 E. $y = 2(x - 1)(x + 5)$

Label 46
 46 If $2\sin 2\theta - \sin \theta \cos \theta - \cos^2 \theta = 0$, the $\tan \theta$

- A. 1 or $\frac{1}{2}$

- B. -1 or $\frac{1}{2}$
 C. 1 or $-\frac{1}{2}$
 D. -1 or $-\frac{1}{2}$
 E. 1 or -2

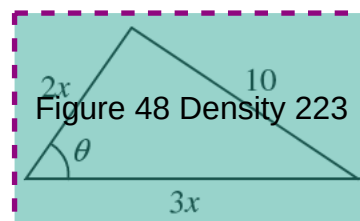
Label 47
 47



In the figure, $VABCD$ is a right pyramid of height 3 cm. The base $ABCD$ is a square of side 2 cm. Let θ be the angle between the face VBC and the base. Find $\tan \theta$.

- A. 1
 B. $\sqrt{2}$
 C. $\frac{3}{2}$
 D. $\frac{3\sqrt{2}}{2}$
 E. 3

Label 48
 48



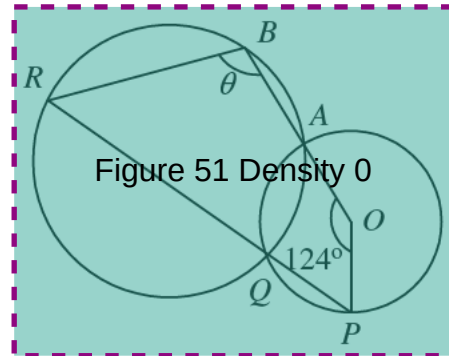
In the figure, if $\cos \theta = \frac{3}{4}$, find the value of x .

- A. 2

- $$E = \frac{\cos \alpha}{\cos \beta}$$

Figure 49a Density 711

Label 51



A. 56°
B. 108°
C. 112°
D. 118°
E. 124°

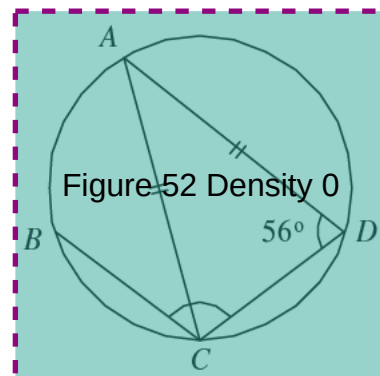
Figure 49b Dens

A	$\frac{5}{2} \text{ m}^2$
B	$5\sqrt{2} \text{ m}^2$
C	$5\sqrt{3} \text{ m}^2$
D	10 m^2
E	

Figure 50a Density 329

A trapezoid with vertices labeled A , B , C , and D . The bottom base is AB and the top base is DC . Arrows on AB and DC indicate they are parallel. The left side is AD and the right side is BC . The interior angle at vertex A is labeled α and the interior angle at vertex B is labeled β .

Label 52



A. 84°
B. 90°
C. 96°
D. 112°
E. 124°

Figure 5b: Density

A.	$\frac{\sin \beta}{\sin \alpha}$
B.	$\frac{\sin \alpha}{\sin \beta}$
C.	$\sin \alpha \sin \beta$
D.	$\frac{\cos \beta}{\cos \alpha}$

Label 53

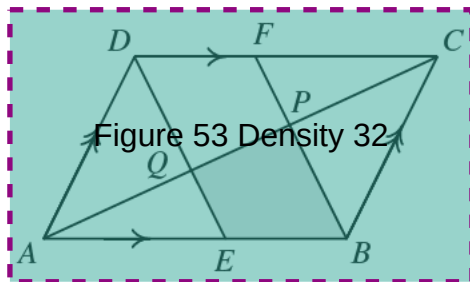


Figure 53 Density 32

In the figure, $ABCD$ is a parallelogram, E and F are the mid-points of AB and DC respectively. BF and ED cut AC at P and Q respectively. If the area of $ABCD$ is 48, find the area of the shaded part.

- A. 6
- B. 8
- C. 9.6
- D. 12
- E. 16

Label 54

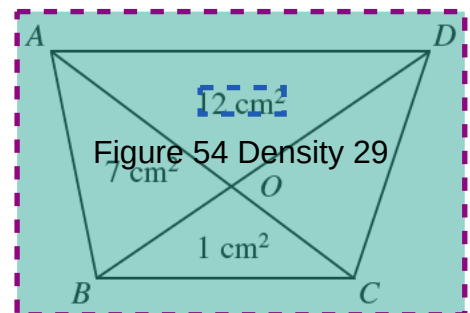


Figure 54 Density 29

In the figure, AC cuts BD at O . The areas of $\triangle AOB$, $\triangle AOD$ and $\triangle BOC$ are 7 cm^2 , 12 cm^2 and 10.5 cm^2 respectively. Find the area of $\triangle OCD$.

- A. 5.5 cm^2
- B. 8 cm^2
- C. 8.5 cm^2
- D. 15.5 cm^2
- E. 18 cm^2